JS_DAY_4_HANDS_ON_GOPIKRISHNA

JS_DAY_4_HANDS_ON_PROBLEM_1_GOPIKRISHNA

1.Assessment Problem_1:

Hands-on Tasks:

- **Add viewport meta tag to the HTML page**
- **Use media queries to:**
  - **Change background color on mobile screen**
  - **Adjust font size for smaller screens**
- **Convert navigation into vertical layout on mobile**
- **Test the page using browser responsive mode**

2.Actual Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Responsive Webpage</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>Responsive Webpage</h1>
  </header>

  <nav>
    <ul>
      <li>Home</li>
      <li>Products</li>
      <li>Contact</li>
    </ul>
  </nav>

  <main>
    <p>
      By reducing the size of the browser window from desktop to mobile view, you can see the background color and
      size of the text changes.
    </p>
  </main>

</body>
</html>
```

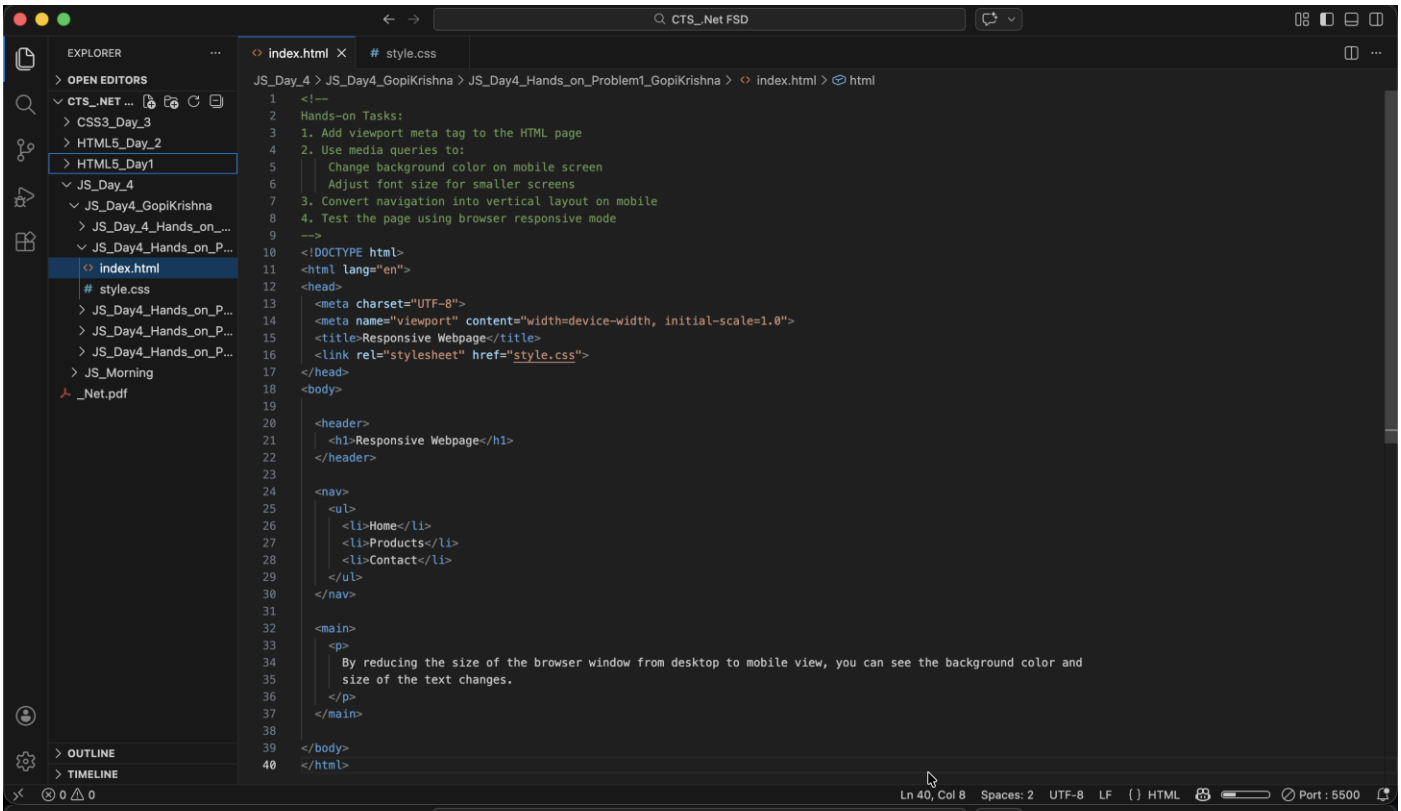
style.css

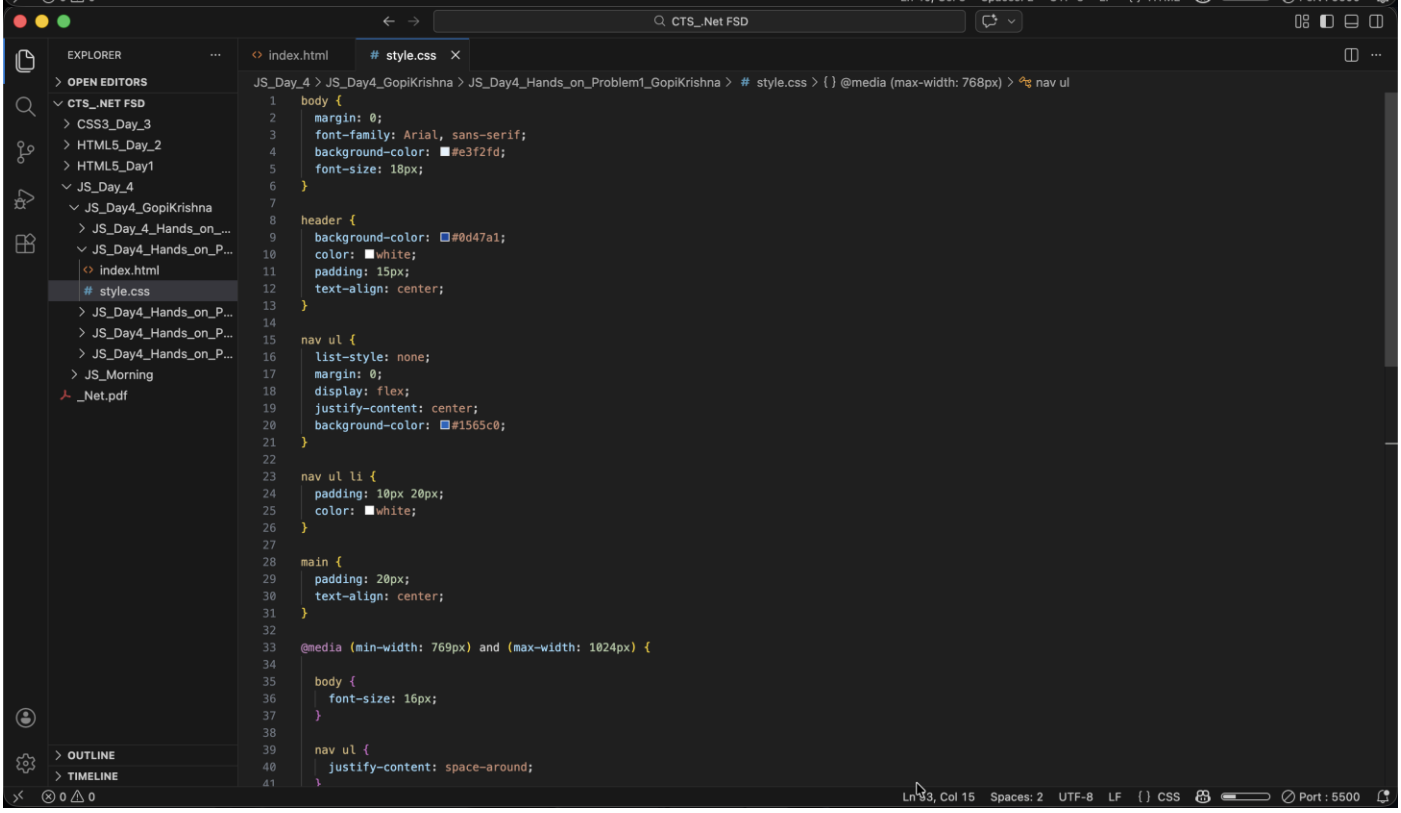
```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #e3f2fd;
```

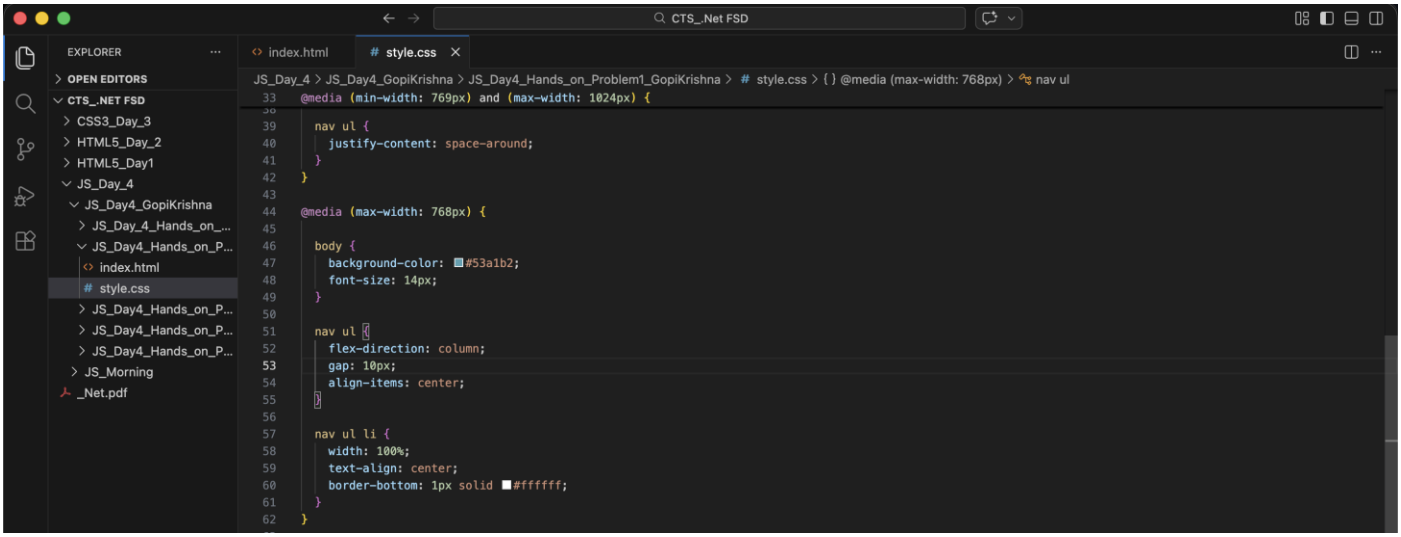
```
  font-size: 18px;
}

header {
  background-color: #0d47a1;
  color: white;
  padding: 15px;
  text-align: center;
}

nav ul {
  list-style: none;
  margin: 0;
  display: flex;
  justify-content: center;
  background-color: #1565c0;
}

nav ul li {
  padding: 10px 20px;
  color: white;
}

main {
  padding: 20px;
  text-align: center;
}

@media (min-width: 769px) and (max-width: 1024px) {

  body {
    font-size: 16px;
  }

  nav ul {
    justify-content: space-around;
  }
}

@media (max-width: 768px) {

  body {
    background-color: #53a1b2;
    font-size: 14px;
  }

  nav ul {
    flex-direction: column;
    gap: 10px;
    align-items: center;
  }

  nav ul li {
    width: 100%;
    text-align: center;
```

```
    border-bottom: 1px solid #ffffff;
  }
}
```

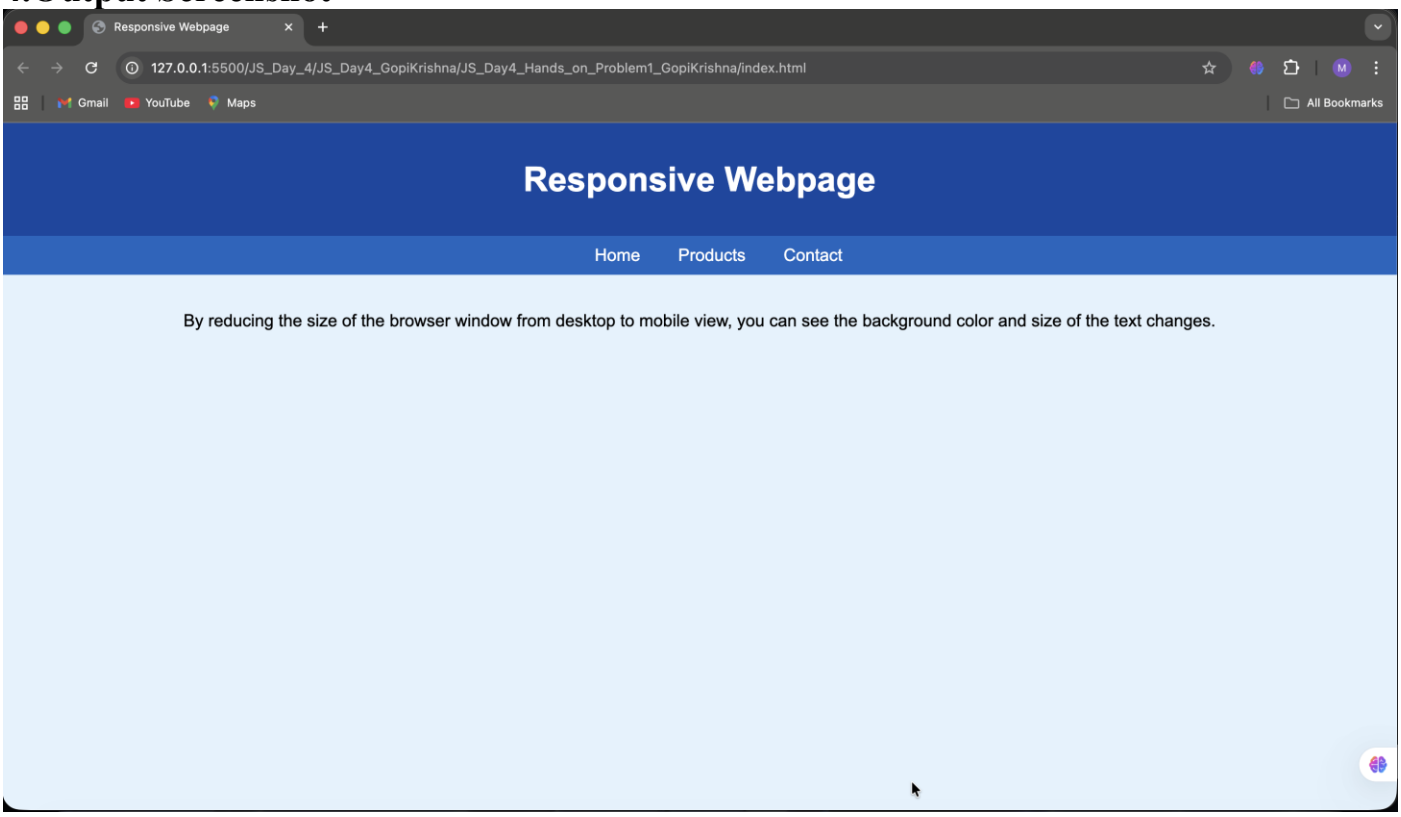
3.Code Screenshot







4.Output Screenshot



5.Explanation

This webpage implements responsive web design using the viewport meta tag and CSS media queries to adapt layout and styling across devices. Different breakpoints modify background color and font size for readability, while the navigation bar transitions from a horizontal Flexbox layout on desktop to a vertical stack on mobile screens. Browser responsive mode is used to validate consistent usability on desktop, tablet, and mobile viewports.

JS_DAY_4_HANDS_ON_PROBLEM_2_GOPIKRISHNA

1. Assessment Problem_2:

Requirements

- **Accept the student's marks as a variable**
- **Use if–else statements to assign grades:**
- **Marks ≥ 75 → Grade A**
- **Marks ≥ 60 → Grade B**
- **Marks ≥ 40 → Grade C**
- **Marks < 40 → Fail**

Display the grade on the web page or console

2. Actual Code:

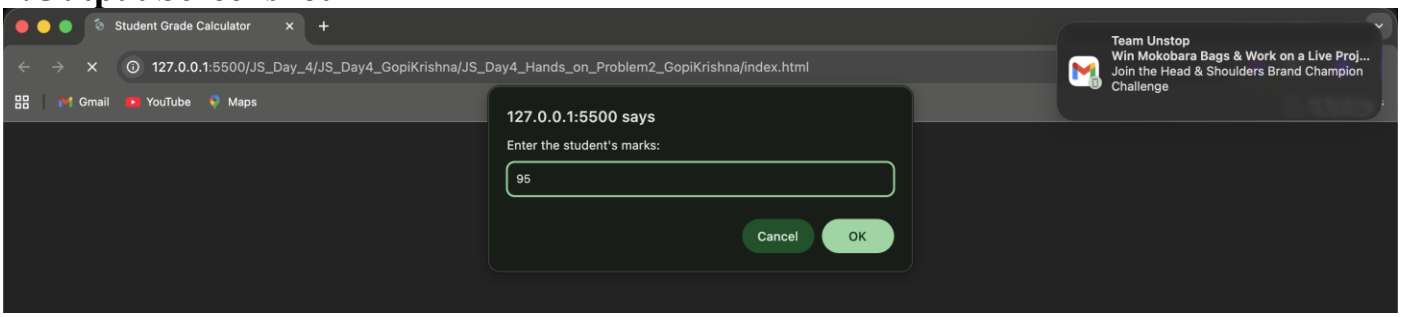
index.html

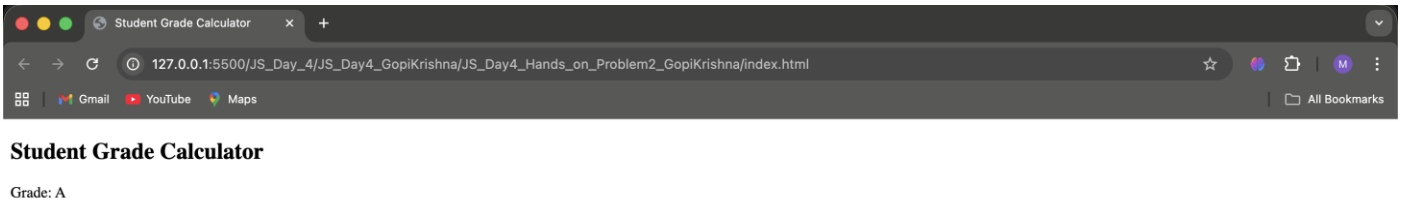
```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Student Grade Calculator</title>
</head>
<body>
   <h2>Student Grade Calculator</h2>
   <script>
      let marks =prompt("Enter the student's marks:");
      if (isNaN(marks)) {
         document.write("Please enter numbers only");
      } else if(marks > 100 || marks < 0){
         document.write("invalid marks");
      } else if(marks <= 100 && marks >= 75){
         document.write("Grade: A");
      } else if(marks >= 60){
         document.write("Grade: B");
      } else if(marks >= 40){
         document.write("Grade: C");
      } else {
         document.write("Grade: Fail");
      }
   </script>
</body>
</html>
```

3. Code Screenshots

```html
<!--
📌 Requirements
    •   Accept the student's marks as a variable
    •   Use if-else statements to assign grades:
    •   Marks ≥ 75 → Grade A
    •   Marks ≥ 60 → Grade B
    •   Marks ≥ 40 → Grade C
    •   Marks < 40 → Fail
Display the grade on the web page or console
-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Grade Calculator</title>
</head>
<body>
    <h2>Student Grade Calculator</h2>
    <script>
        let marks =prompt("Enter the student's marks:");
        if (isNaN(marks)) {
            document.write("Please enter numbers only");
        } else if(marks > 100 || marks < 0){
            document.write("invalid marks");
        } else if(marks <= 100 && marks >= 75){
            document.write("Grade: A");
        } else if(marks >= 60){
            document.write("Grade: B");
        } else if(marks >= 40){
            document.write("Grade: C");
        } else {
            document.write("Grade: Fail");
        }
    </script>
</body>
</html>
```

4.Output Screenshot





5. Explanation

This program implements a basic decision-making workflow in JavaScript by capturing numeric input using prompt(), storing it in a variable declared with let, and validating it using isNaN() and range checks. Conditional logic is applied through an if–else if–else chain with comparison and logical operators to evaluate the marks against predefined thresholds and assign the appropriate grade. The computed result is rendered directly to the webpage using document.write(), demonstrating variable usage, numeric comparisons, and control flow without relying on functions or arrays.

JS_DAY_4_HANDS_ON_PROBLEM_3_GOPIKRISHNA

1. Assessment Problem_3:

Requirements

- Store purchase amount in a variable
- Apply discount rules:
    - Amount ≥ 5000 → 20% discount
    - Amount ≥ 3000 → 10% discount
    - Amount < 3000 → No discount
- Calculate and display:
    - Discount amount
    - Final payable amount

2. Actual Code:

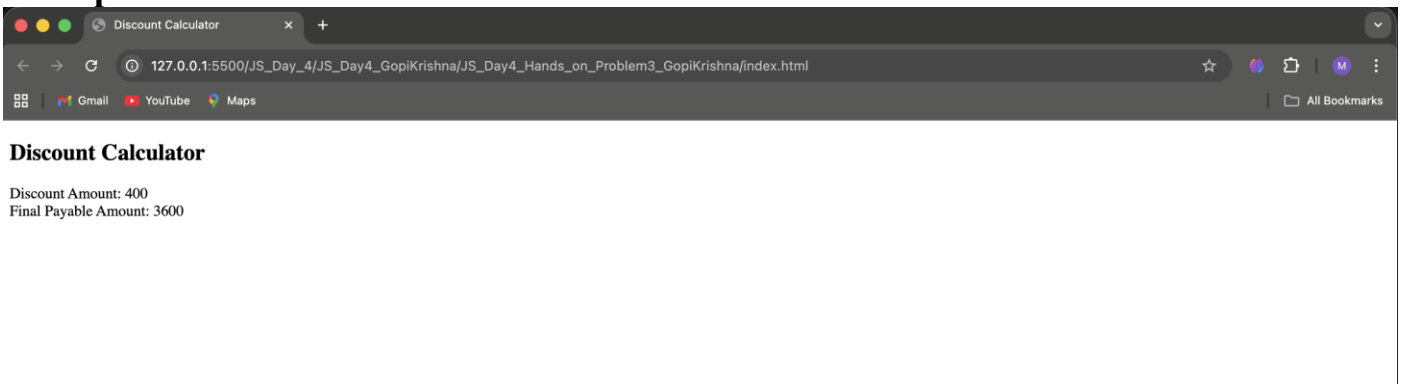
index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Discount Calculator</title>
</head>
<body>
   <h2>Discount Calculator</h2>

   <script>
      let totalamount=prompt("Enter the total purchase amount:");
      if (isNaN(totalamount)) {
         document.write("Please enter numbers only");
      } else if(totalamount < 0){
         document.write("invalid amount");
      } else if(totalamount >= 5000){
         let discountamount = totalamount * 0.20;
         let finalamount = totalamount - discountamount;
         document.write("Discount Amount: " + discountamount + "<br>");
         document.write("Final Payable Amount: " + finalamount);
      } else if(totalamount >= 3000){
         let discountamount = totalamount * 0.10;
         let finalamount = totalamount - discountamount;
         document.write("Discount Amount: " + discountamount + "<br>");
         document.write("Final Payable Amount: " + finalamount);
      } else {
         document.write("No discount applied. Final Payable Amount: " + totalamount);
      }
   </script>
</body>
</html>
```

3. Code Screenshots

```html
    Store purchase amount in a variable
    Apply discount rules:
    •   Amount ≥ 5000 → 20% discount
    •   Amount ≥ 3000 → 10% discount
    •   Amount < 3000 → No discount
    Calculate and display:
    •   Discount amount
    •   Final payable amount
-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Discount Calculator</title>
</head>
<body>
    <h2>Discount Calculator</h2>

    <script>
        let totalamount=4000;
        if (isNaN(totalamount)) {
            document.write("Please enter numbers only");
        } else if(totalamount < 0){
            document.write("invalid amount");
        } else if(totalamount >= 5000){
            let discountamount = totalamount * 0.20;
            let finalamount = totalamount - discountamount;
            document.write("Discount Amount: " + discountamount + "<br>");
            document.write("Final Payable Amount: " + finalamount);
        } else if(totalamount >= 3000){
            let discountamount = totalamount * 0.10;
            let finalamount = totalamount - discountamount;
            document.write("Discount Amount: " + discountamount + "<br>");
            document.write("Final Payable Amount: " + finalamount);
        } else {
            document.write("No discount applied. Final Payable Amount: " + totalamount);
        }
    </script>
</body>
</html>
```

4.Output Screenshot



5. Explanation

This program demonstrates conditional business logic implementation in JavaScript by storing a hardcoded purchase amount in a numeric variable and applying arithmetic operations to compute discounts based on predefined thresholds. An if–else if–else control structure evaluates the amount and applies the appropriate discount rate, calculates the discount value, and derives the final payable amount using basic operators. The results are rendered directly to the webpage using document.write(), reinforcing the use of primitive data types, expressions, and decision-making without user input or advanced language constructs.

JS_DAY_4_HANDS_ON_PROBLEM_4_GOPIKRISHNA

1. Assessment Problem_4:

Requirements

- Store signal color in a variable ("red", "yellow", "green")
- Use a **switch statement** to display:
- Red → Stop

- Yellow → Get Ready
- Green → Go

Handle invalid signal input gracefully

2. Actual Code:

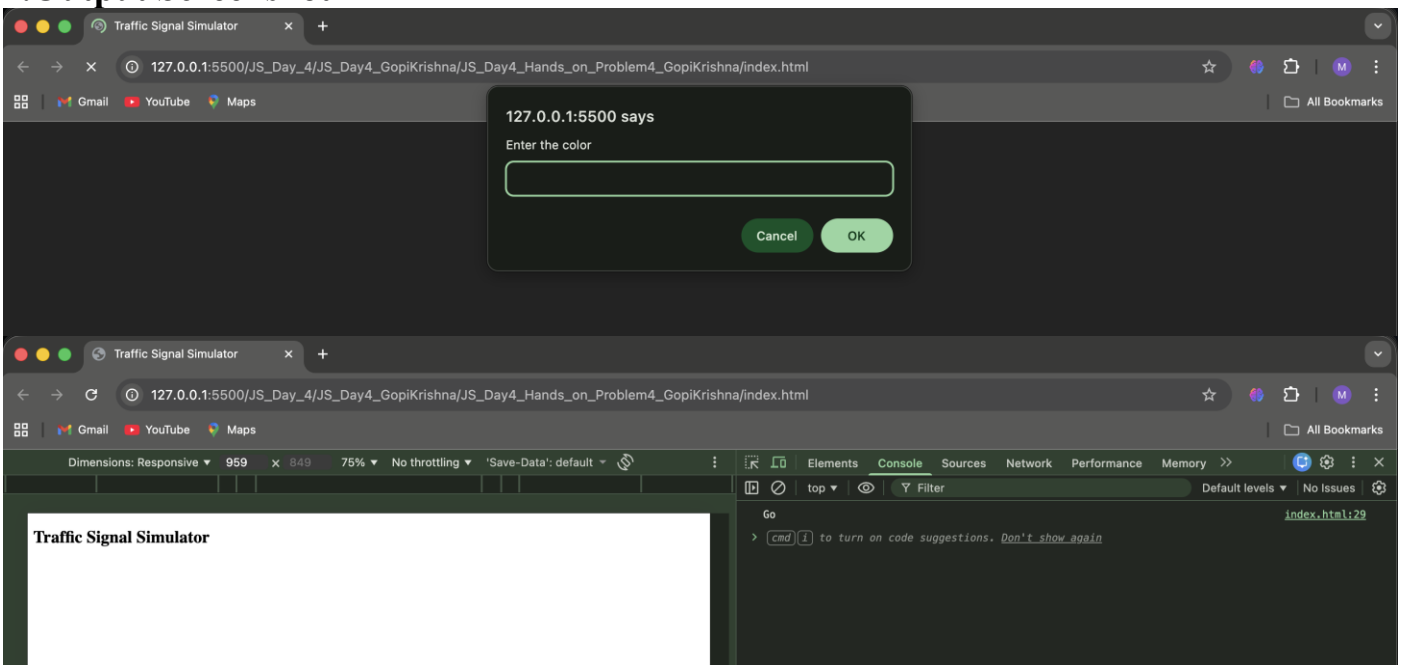
index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Traffic Signal Simulator</title>
</head>
<body>
   <h2>Traffic Signal Simulator</h2>
   <script>
      let signalColor = prompt("Enter the color").toLowerCase();
      switch (signalColor) {
         case "red":
            console.log("Stop");
            break;
         case "yellow":
            console.log("Get Ready");
            break;
         case "green":
            console.log("Go");
            break;
         default:
            console.log("Invalid signal color, please enter red, yellow, or green.");
      }
   </script>
</body>
</html>
```

3. Code Screenshots

```html
<!--JS_Day4_Hands_on_Problem4_GopiKrishna
Requirements
    •   Store signal color in a variable ("red", "yellow", "green")
    •   Use a switch statement to display:
    •   Red → Stop
    •   Yellow → Get Ready
    •   Green → Go
Handle invalid signal input gracefully
-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Traffic Signal Simulator</title>
</head>
<body>
    <h2>Traffic Signal Simulator</h2>
    <script>
        let signalColor = prompt("Enter the color").toLowerCase();
        switch (signalColor) {
            case "red":
                console.log("Stop");
                break;
            case "yellow":
                console.log("Get Ready");
                break;
            case "green":
                console.log("Go");
                break;
            default:
                console.log("Invalid signal color, please enter red, yellow, or green.");
        }
    </script>
</body>
</html>
```

4.Output Screenshot



5. Explanation

This program implements a switch-based control flow mechanism in JavaScript by storing the traffic signal state as a string variable and normalizing input using toLowerCase() for consistent comparison. A switch–case statement maps each valid signal value (red, yellow, green) to its corresponding instruction and outputs the result using console.log(). The default case handles invalid inputs gracefully, demonstrating clean multi-branch decision-making without relying on if–else, and reinforcing structured control flow and string comparison concepts.

JS_DAY_4_HANDS_ON_PROBLEM_5_GOPIKRISHNA

1. Assessment Problem_4:

Requirements

- Store a number in a variable
- Use conditional (ternary) operator to check:
- Positive or Negative
- Use if–else to check:
- Even or Odd
- Use a loop to print all numbers from 1 to the given number

2. Actual Code:

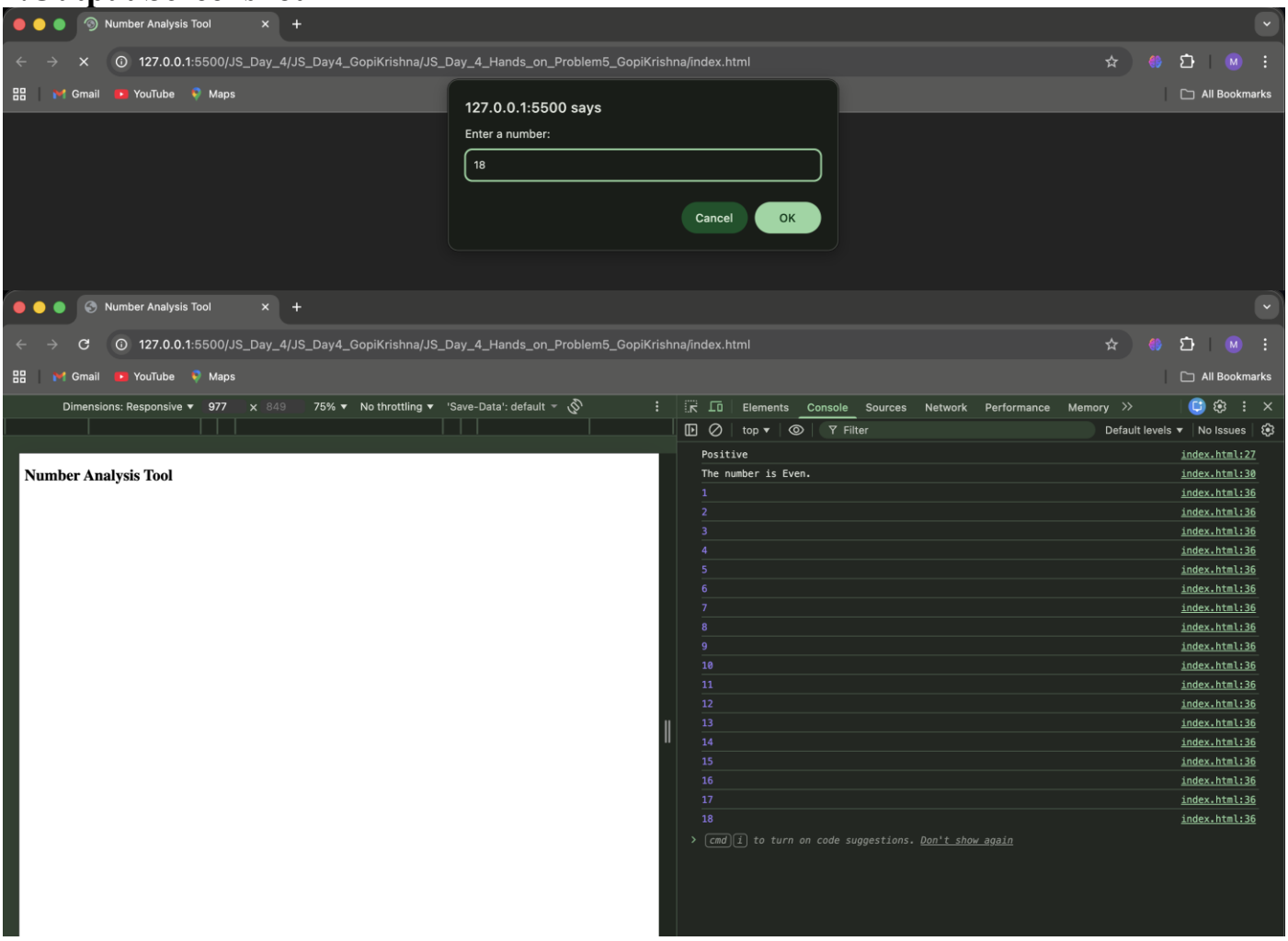
index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Number Analysis Tool</title>
</head>
<body>

   <h2>Number Analysis Tool</h2>
   <script>
      let number = prompt("Enter a number:");
      if(isNaN(number)){
         console.log("Please enter a valid number.");
      } else {
         number = parseInt(number);
         let result = number == 0 ? "Zero" : number > 0 ? "Positive" : "Negative";
         console.log(result);

         if(number % 2 == 0){
            console.log("The number is Even.");
         } else{
         console.log("The number is Odd.");
      }

      for(let i=1;i<=number;i++){
         console.log(i);
      }
    }

   </script>
</body>
</html>
```

3. Code Screenshots

```html
Requirements
    •   Store a number in a variable
    •   Use conditional (ternary) operator to check:
    •   Positive or Negative
    •   Use if-else to check:
    •   Even or Odd
    •   Use a loop to print all numbers from 1 to the given number
-->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Number Analysis Tool</title>
</head>
<body>

    <h2>Number Analysis Tool</h2>
    <script>
        let number = prompt("Enter a number:");
        if(isNaN(number)){
            console.log("Please enter a valid number.");
        } else {
            number = parseInt(number);
            let result = number == 0 ? "Zero" : number > 0 ? "Positive" : "Negative";
            console.log(result);

            if(number % 2 == 0){
                console.log("The number is Even.");
            } else{
            console.log("The number is Odd.");
        }

        for(let i=1;i<=number;i++){
            console.log(i);
        }
    }
    </script>
</body>
</html>
```

4.Output Screenshot



5. Explanation

This program implements a multi-step numerical analysis workflow in JavaScript by accepting a numeric value, validating and parsing it, and applying different control flow constructs for analysis. A conditional

(ternary) operator determines whether the number is positive, negative, or zero, while an if–else statement evaluates its parity using the modulus operator. A for loop then iterates from 1 to the given number and outputs each value to the console, demonstrating the combined use of variables, conditional operators, decision statements, and iterative control flow to solve a composite logic problem.